Step 2a - H1 - Pharma continuous solvent removal

Market scan

Market size

The parent market for this product idea is pharmaceutical continuous-manufacturing equipment, estimated at ~$2.7B in 2025 and growing at ~10-11% CAGR toward ~$4.6B by 2030; the broader pharma drying-equipment market is ~$5.8B (2024) heading to ~$8.2B (2029). The solvent-removal slice is a niche within these: organic-solvent granulation and coating have been a minority of oral-solid-dose processes since aqueous systems largely displaced them in the 1990s, and persist mainly for moisture-sensitive APIs, functional/enteric coatings, effervescents and contained HPAPI work. Order of magnitude for the addressable niche (continuous-capable, solvent-rated drying/coating equipment): ~$0.3-0.8B globally.

Region	Size ($B)	Region	Size ($B)
Europe	~$1.0B (largest continuous-manufacturing region in 2025, Western Europe; dense GEA/Glatt/Bohle/Syntegon install base)	Asia	~$0.7B (fastest-growing region; India/China OSD capacity)
America	~$0.9B (US-led; strong CM adoption push by FDA)	Africa	~<$0.1B (minimal pharma solids manufacturing)

Target customers

Buyers are solid-dose manufacturers that run or want solvent-based wet granulation/coating continuously: a subset of the ~7,100 FDA-registered drug-product manufacturing sites (~4,000+ companies, 79 countries) plus the ~400-750 solid-dose-capable CDMOs. The realistic solvent-process subset is a minority of these, concentrated among large pharma, larger CDMOs and mid-size generics in the US, EU and India. Reach is hard: capital equipment sold through long (12-36 month) pharma procurement cycles, validation burden, conservative process-engineering buyers, and sales require application-lab proof per formulation.

Customer size	Approx. number	Market share (in %)
Large	~100-200 companies (top-50 pharma + large CDMOs like Lonza/Catalent/Thermo Patheon class)	~70% of relevant equipment spend
Medium	~500-1,000 companies (mid-size generics, regional CDMOs, specialty pharma)	~20-25%
Small	~1,000-3,000 companies (small generics, nutraceutical; only a minority run solvent processes)	~5-10%

Players offering a solution today

Roughly 10-15 credible equipment OEMs serve pharma continuous solids processing; no single dominant vendor but GEA and Glatt are the reference leaders for continuous granulation/drying lines, with Syntegon, L.B. Bohle, Romaco, ACG and Fette covering most remaining share. Nobody today sells a continuous microwave-based solvent-removal module for pharma; microwave-vacuum drying exists only in batch

single-pot processors (GEA UltimaPro, Bohle VMA) and in non-pharma industrial MW dryers.

Player	Type of solution	Market share (in %)
Large (GEA, Glatt, Syntegon, Romaco, ACG, Fette Compacting)	Integrated continuous lines (GEA ConsiGma, Glatt MODCOS) with twin-screw granulation + segmented fluid-bed drying + coating; single-pot processors; batch fluid beds and drum coaters	~60-70% combined of pharma continuous solids equipment
Medium (L.B. Bohle, Freund-Vector, O'Hara, Driam, Diosna, IMA, Coperion K-Tron)	Continuous granulation/drying modules (QbCon), continuous coaters (Driaconti), lab/pilot lines, feeders	~20-30%
Small (Pueschner, SAIREM, Kerone, Microdry + Chinese/Indian OEMs)	Industrial microwave drying systems (non-pharma focus), low-cost batch dryers and coaters	~10%

Regulation

Regulation	Jurisdiction	What it means for us
ICH Q13 Continuous Manufacturing	Global (ICH regions incl. FDA, EMA, PMDA)	Regulatory pathway for CM now harmonized since Nov 2022 - supports our pitch that regulators accept continuous solvent-removal unit operations; enables batch-to-continuous conversion
EU GMP / FDA cGMP (21 CFR 210/211) + 21 CFR Part 11 / Annex 11	EU / US	Equipment must be GMP-cleanable, validatable, with compliant data systems - raises development cost but also barriers protecting incumbents; we must design for PAT and cleaning validation
ATEX Directive 2014/34/EU (+NEC/NFPA in US)	EU / US	Organic-solvent vapours require explosion-rated equipment, inerting (N2) and LEL monitoring; a microwave applicator inside a solvent atmosphere is a real engineering and certification challenge
ICH Q3C Residual Solvents	Global	Defines acceptable solvent levels in drug product; drying endpoint must be demonstrably controlled - argues for in-line moisture PAT

EU Industrial Emissions / VOC rules + solvent recovery expectation	EU	Solvent vapour cannot be vented; condensation/recovery loop required - actually favors closed MW-vacuum systems over high-airflow convective dryers
Machinery Directive 2006/42/EC, CE marking	EU	Standard machine safety; microwave leakage limits (~5 mW/cm2) must be engineered into a continuous open-ended applicator

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis